

Lesson Exemplar in General Science

Quarter 3

LE

6

GOVERNMENT PROPERTY
NOT FOR SALE

Lesson Exemplar for General Science
Quarter 3: Unit 6

This material is intended exclusively for the use of senior high school teachers participating in the implementation of the Strengthened Senior High School Curriculum. It aims to assist in delivering the curriculum content, standards, and lesson competencies. Any unauthorized reproduction, distribution, modification, or utilization of this material beyond the designated scope is strictly prohibited and may result in appropriate legal actions and disciplinary measures.

Borrowed content included in this material are owned by their respective copyright holders. Every effort has been made to locate and obtain permission to use these materials from their respective copyright owners. The publisher and development team do not represent nor claim ownership over them.

Development Team

Writer: Bryan R. Concepcion, RN
Region II
Division of Isabela

Validator: Amy Janine V. Gay D. Erguiza

Language Editor:

Consultant: Crist John Pastor, PhD
Philippine Normal University

Learning Area Specialist: John Cyrus L. Doblada

Bureau of Learning Delivery
Bureau of Curriculum Development

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph.

LESSON EXEMPLAR

Learning Area	General Science	Grade Level	Grade 11
Semester	2 nd Semester	Quarter	Quarter 3

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	The learners learn that: Organ systems work together to carry out essential functions of an organism.
Performance Standard	<i>By the end of the quarter</i>, learners use examples to describe the relationships among living systems and their components, and how the structure of a part relates to its function. They explain the role of cells in perpetuating life and how organ systems work together to maintain optimal health. They conduct investigations on the effects of climate change on local ecosystems.
Learning Competencies	Key Topics Learned in the Previous Grade Level In this topic, the learners are assumed to have prior knowledge of the following topics: - Overview of the Nervous System and Endocrine System - Integration of Nervous and Endocrine Systems - Homeostasis and feedback mechanisms The learners: Explains how the nervous and endocrine systems work together to regulate body temperature in response to stimuli.
II. REFERENCES and MATERIALS <i>(Selecting Resources and Materials)</i>	• References: -Khan Academy. “Temperature Regulation Strategies.” Accessed October 18, 2025. https://www.khanacademy.org/science/biology/principles-of-physiology/metabolism-and-thermoregulation/a/animal-temperature-regulation-strategies.. -Open Textbooks for Hong Kong. “Thermoregulation.” Last modified April 6, 2016. https://www.opentextbooks.org.hk/ditatopic/35173.. -Cherry, Kendra. “Nervous and Endocrine Systems: How They Interact and Impact Behavior.” Verywell Mind. Last updated November 28, 2023. https://www.verywellmind.com/the-nervous-and-endocrine-systems-2794894..

	-YouTube. “Student Athletes Susceptible to Heat Exhaustion and Heat Stroke.” Video, 2:30. Posted by WXYZ-TV Detroit, August 1, 2023. https://www.youtube.com/watch?v=DEdt6KGSJ4g.. -Homework Clinic. “The Hypothalamus: The Body’s Thermostat (Human Thermostat).” YouTube video, 3:45. Posted March 15, 2022. https://www.youtube.com/watch?v=ac_8bjEZaVE.. -Longman, Jeré. “Athletes Suffer as Heat Rises.” New York Times, May 5, 2012. https://www.nytimes.com/2012/05/05/sports/05iht-athlete05.html.. -Choi, Charles Q. “Why Do We Shiver When We’re Cold?” Live Science. Last updated July 22, 2022. https://www.livescience.com/32475-why-do-we-shiver-when-cold.html.. -Goldstein, David S. “Adrenal Responses to Stress.” Journal of Clinical Investigation 130, no. 2 (2020): 559–561. https://doi.org/10.1172/JCI135006.. • Flow charts, scripts, and visual aids • Online quiz tools (Kahoot, Quizizz, Google Forms, or any available online quiz platform) • Slide Decks or any Interactive digital notebooks, PhET • Biology textbook, curriculum guide, teacher’s guide
<i>(These shall be accomplished per topic)</i>	
III. CONTENT <i>(Sequencing Content)</i>	Organ Systems Working Together • Effectors in Thermoregulation (Sweat glands, muscles, and blood vessels) • Responses of the Body to Temperature Changes (Observable actions such as shivering, sweating, vasodilation, and vasoconstriction.) • Hormonal Regulation in Thermoregulation • The Hypothalamus as the Control Center of Thermoregulation • Integration of the Nervous and Endocrine Systems (Neuroendocrine) in Thermoregulation
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	➤ Describe the role of the hypothalamus in detecting temperature changes and coordinating responses between the nervous and endocrine systems. ➤ Explain how the nervous system initiates immediate thermoregulatory actions such as sweating and shivering in response to heat and cold stimuli. ➤ Identify key endocrine glands (e.g., thyroid gland) and their hormonal contributions to long-term temperature regulation.

	➤ Analyze a real-life scenario to trace how both systems interact to maintain homeostasis during temperature fluctuation.	
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lessons and Assessing Learning)</i> This section focuses on selecting learner-centered, evidence-based instructional approaches such as problem-based learning, collaborative tasks, interdisciplinary integration, and technology-enhanced instruction. These strategies are intended to foster active engagement, critical thinking, and adaptability across diverse learning pathways. The chosen approaches and methodologies will be reflected through varied and relevant activities and assessments that emphasize real-world relevance and application, thereby enhancing learner engagement and comprehension.		ANNOTATION <i>*Instructions to the teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge	1. Activating Prior Knowledge Option 1: Think-Pair-Share Prompt Question: <i>“What happens when you feel too hot or too cold?”</i> Procedure: - The teacher asks students to individually write down what they notice in their bodies when they are exposed to hot or cold environments. - Students pair up and share their answers. - Each pair contributes one idea to a class discussion (e.g., sweating, shivering, goosebumps).	IDF Relevance: These three activities, Think-Pair-Share, Picture-Based Prompt, and Mini Quiz Game, are aligned with the DepEd Instructional Design Framework. The Think-Pair-Share activity builds contextual relevance by prompting students to reflect on personal experiences of heat and cold, fostering a connection between lived sensations and scientific concepts. The Picture-Based Prompt encourages creativity and hypothesis-building as students interpret visual cues and predict internal body responses, laying the groundwork for conceptual understanding. The Mini Quiz Game

Expected Output: The students will have a list of observable body responses to heat/cold.

Option 2. Picture-based Prompt

Procedure:

- Show a contrasting image: a sweating athlete, a shivering mountaineer.
- Students describe what they think is happening inside the body.
- The teacher records student ideas on the board to compare with actual thermoregulation processes later.

Expected Output: Students' hypotheses on body regulation.

Option 3: Mini Quiz Game

Procedure:

- Teacher reads out 10 True/False statements
 - ❖ Here are some sample questions:
 1. *The nervous system uses electrical signals to transmit messages throughout the body.*
 2. *The nervous system releases hormones to control bodily processes.*
 3. *The brain region known as the hypothalamus aids in controlling body temperature.*
 4. *The endocrine system regulates the response of sweating.*
 5. *Compared to the nervous system, the endocrine system reacts to stimuli more quickly.*
 6. *Stress causes the release of the hormone adrenaline, which aids in controlling body temperature.*
 7. *When it's too hot, shivering helps the body cool down.*



Source:
<https://www.nytimes.com/2012/05/05/sports/05iht-athlete05.html>



Source:
<https://www.livescience.com/32475-why-do-we-shiver-when-cold.html>

promotes collaboration and active engagement through peer interaction and immediate feedback, while also serving as a diagnostic tool to assess prior knowledge. Pedagogically, these activities support constructivist and formative approaches by activating prior experiences, encouraging peer dialogue, and scaffolding learning through visual, verbal, and kinesthetic strategies.

Note for Teacher:

The activity is a low-risk, high-engagement way to start the lesson. During the activity, go around the room to listen for misconceptions of the students (e.g., “sweating cools you down because your body releases heat directly”) and use these insights to guide the class discussion. When collecting ideas from the class, cluster responses into categories, nervous vs. endocrine responses, to begin building conceptual connections.

During the activity, you may also use the moment to introduce key vocabulary informally, such as vasodilation and hypothalamus, as a preview for the next part of instruction. Make the activity as engaging as possible.

	8. <i>The thyroid gland aids in controlling body temperature and metabolism.</i> 9. <i>The endocrine and nervous systems do not interact with each other, and they function independently.</i> 10. <i>Goosebumps are a response to cold controlled by the nervous system</i> <i>Answer Key: T-F-T-F-F-T-F-T-F-T</i> - Students raise a red card for False, green card for True. - The teacher provides immediate feedback to the responses of the students. Expected Output: Quick diagnostic of prior knowledge. 2. Establishing the Purpose of the Lesson Option 1. Body System Simulation (Inside the Thermostat) General Directions: - This activity uses a role-based simulation to help students visualize and embody the internal coordination between the nervous and endocrine systems during thermoregulation. Students are assigned specific roles and act out the sequence of biological responses to a temperature stimulus. The teacher facilitates the simulation and leads a debriefing session to reinforce the specific concepts. Materials: - Role cards (hypothalamus, sensory neurons, motor neurons, sweat glands, muscles, endocrine glands) - Scenario cards (e.g., heatwave, icy wind) - Whiteboard and whiteboard marker for diagramming Procedure: - Assign each student a role from the thermoregulation system. - Present a temperature scenario (e.g., sudden heatwave or icy wind) - Students act out the biological response:	IDF Relevance: The three given activities, Body System Simulation, Thermoregulation Relay, and Comic Strip Creation, strongly align with the DepEd Instructional Design Framework by integrating the four key aspects: context, connection, collaboration, and creativity. The Body System Simulation provides contextual learning by immersing students in role-play scenarios that mirror real physiological processes, fostering a connection between abstract concepts and bodily functions. The Thermoregulation Relay promotes collaboration and active engagement through team-based challenges that reinforce system responses to heat and cold. Comic Strip Creation encourages creativity and personal expression as students visualize and narrate thermoregulation using
--	--	---

	▪ The sensory neurons detect the stimulus and send signals to the hypothalamus. ▪ The hypothalamus interprets the signal and decides the appropriate response. ▪ The motor neurons activate the sweat glands or muscles. ▪ The endocrine glands release hormones for long-term regulation. - Debrief with a class discussion and diagram linking each role to its biological function. <i>Thermoregulation Response Flow Diagram</i> [Environmental Stimulus] ↓ (Heatwave or Icy Wind) ↓ [Sensory Neurons] → Detect temperature change → Send signal to hypothalamus ↓ [Hypothalamus] → Interprets signal → Decides appropriate response ↓ <table border="0" style="width: 100%;"><tr><td style="width: 33%; vertical-align: top;">[Motor Neurons] → Activate: - Sweat glands (heat) - Muscles for shivering (cold) ↓</td><td style="width: 33%; vertical-align: top;">[Endocrine Glands] → Release hormones (e.g., adrenaline, TSH)</td><td style="width: 33%; vertical-align: top;">[Behavioral Response] → Seek shade, add layers</td></tr></table> [Effector Organs] → Sweat glands produce sweat (cooling) → Muscles contract (shivering for warmth)	[Motor Neurons] → Activate: - Sweat glands (heat) - Muscles for shivering (cold) ↓	[Endocrine Glands] → Release hormones (e.g., adrenaline, TSH)	[Behavioral Response] → Seek shade, add layers	scientific accuracy and imaginative storytelling. Pedagogically, these activities support experiential, cooperative, and differentiated learning approaches, making science concepts accessible, memorable, and meaningful for all learners. Option 1 Note for Teacher: The teacher may use the provided diagram as a foundational visual aid during the debriefing session to illustrate the coordination between the hypothalamus, nervous system, and endocrine system in thermoregulation. They are encouraged to adapt, expand, or contextualize the diagram based on the students' responses, grade level, or specific learning needs. Additional elements, such as behavioral responses (e.g., seeking shade or adding clothing), hormone names, or real-life examples, can be incorporated to enrich understanding and reinforce connections to everyday experiences. This flexibility allows the diagram to serve not only as a scientific reference but also as a dynamic tool for discussion and deeper learning.
[Motor Neurons] → Activate: - Sweat glands (heat) - Muscles for shivering (cold) ↓	[Endocrine Glands] → Release hormones (e.g., adrenaline, TSH)	[Behavioral Response] → Seek shade, add layers			

Option 2. Hot vs. Cold Challenge: Thermoregulation Relay

General Directions:

- This activity uses a relay-style competition to reinforce thermoregulation concepts. The students work in teams to complete a series of stations that represent stages in the body's responses to heat or cold. The game format encourages engagement, teamwork, and rapid recall of the key concepts.

Materials:

- Station cards (stimulus, hypothalamus response, nervous system action, endocrine hormone)
- Stopwatch
- Answer sheets

Procedure:

- Divide the class into two teams:
 - Team 1 – Heat Response
 - Team 2 – Cold Response
- The teacher sets up four stations:
 - Station 1: Identify the stimulus (heat or cold).
 - Station 2: Match the hypothalamus response.
 - Station 3: Choose the correct nervous system action (sweating or shivering).
 - Station 4: Identify the endocrine hormone involved.
- The teams race to complete all the stations accurately.
- After the activity, review answers as a class and clarify misconceptions.

Sample Thermoregulation Relay Answer Sheet

Station	Prompt	Team Response
1. Stimulus	What environmental condition triggered the response?	

Option 2 Note for Teacher:

Encourage students to explain their reasoning during the review. Clarify misconceptions, especially around hormone functions. Also, reinforce that the hypothalamus is the central coordination between the nervous and endocrine systems.

	2. Hypothalamus response	What does the hypothalamus detect and decide?		
	3. Nervous system response	What immediate action does the nervous system initiate?		
	4. Endocrine hormone	What hormone is released to support long-term regulation?		
	Heat Response Sample Expected Answers: • Station 1: Exposure to heat or an elevated body temperature • Station 2: Hypothalamus detects overheating and signals a cooling response • Station 3: Activates sweat glands to release sweat. • Station 4: Releases anti-diuretic hormone (ADH) or reduces thyroid hormone activity to conserve energy. Cold Response Sample Expected Answers: • Station 1: Exposure to cold or a drop in body temperature • Station 2: Hypothalamus detects cold and signals a warming response • Station 3: Activates muscles to produce shivering. • Station 4: Releases thyroid hormones to increase metabolic heat production. Option 3. Thermoregulation: A Comic Strip Creation General Directions: - This creative activity allows students to visualize and narrate the thermoregulation process through comic art. By designing a character’s experience with heat or cold, students demonstrate their understanding of how the body detects and responds to temperature changes.			
Option 3 Note for Teacher: Encourage students that their comic must include detection by the hypothalamus, nervous system response (sweating or shivering), endocrine system support				

	Materials: - Comic strip panels template - Art materials - Reference diagram of the nervous and endocrine systems Procedure: - Students choose a scenario: ▪ Scenario 1 – Extreme Heat ▪ Scenario 2 – Extreme Cold - The students create a comic strip showing: ▪ Detection by the hypothalamus ▪ Nervous system response (sweating or shivering) ▪ Endocrine system support - Encourage creativity, humor, and specific accuracy. - After the activity, let the students display their comic strip and let them explain their panels in class. Comic strip panels template What to draw or What to put: <table><tr><th>Panel</th><th>What to draw or What to put:</th><th>Sample Caption Prompt</th></tr><tr><td>1</td><td>Your character is experiencing heat or cold</td><td>“It’s so hot/cold today! My body feels strange...”</td></tr><tr><td>2</td><td>The hypothalamus detects the temperature change</td><td>“The hypothalamus senses a change in body temperature!”</td></tr><tr><td>3</td><td>The nervous system initiates a response</td><td>“Time to sweat/shiver! Let’s cool down/warm up!”</td></tr><tr><td>4</td><td>The endocrine system releases hormones</td><td>“Hormones are on the way to help regulate!”</td></tr><tr><td>5</td><td>Your character is returning to balance</td><td>“Phew! I feel normal again. Thanks, body systems!”</td></tr></table>	Panel	What to draw or What to put:	Sample Caption Prompt	1	Your character is experiencing heat or cold	“It’s so hot/cold today! My body feels strange...”	2	The hypothalamus detects the temperature change	“The hypothalamus senses a change in body temperature!”	3	The nervous system initiates a response	“Time to sweat/shiver! Let’s cool down/warm up!”	4	The endocrine system releases hormones	“Hormones are on the way to help regulate!”	5	Your character is returning to balance	“Phew! I feel normal again. Thanks, body systems!”	(hormone release), and a creative storyline or humorous twist (optional but encouraged)
Panel	What to draw or What to put:	Sample Caption Prompt																		
1	Your character is experiencing heat or cold	“It’s so hot/cold today! My body feels strange...”																		
2	The hypothalamus detects the temperature change	“The hypothalamus senses a change in body temperature!”																		
3	The nervous system initiates a response	“Time to sweat/shiver! Let’s cool down/warm up!”																		
4	The endocrine system releases hormones	“Hormones are on the way to help regulate!”																		
5	Your character is returning to balance	“Phew! I feel normal again. Thanks, body systems!”																		

B. Instituting New Knowledge	1. Presenting Examples Option 1: Thermometer Demonstration General Directions: - This activity uses a hands-on demonstration to help students visualize how the body detects and responds to temperature changes. By observing thermometers placed in hot or cold water, students will draw parallels between external temperature readings and internal sensory detection. Materials: - 2 clear containers (e.g., beakers or cups) - Hot water and cold water - 2 thermometers (digital or analog) - Whiteboard or chart paper - Markers - Reference diagram of the neuroendocrine system (optional) Procedure: - Present and example – Thermometer demonstration ▪ Fill one container with hot water and the other with cold water. ▪ Place a thermometer in each and allow students to observe the readings. ▪ Ask students to describe what they see and compare the temperature changes. - Connecting to thermoregulation ▪ Explain that the thermometers represent sensory receptors in the skin that detect temperature changes. ▪ Ask the students to predict how the body would respond to each condition: ➤ Hot water → sweating, vasodilation ➤ Cold water → shivering, vasoconstriction - Role of the hypothalamus ▪ Introduce the hypothalamus as the body’s control center.	IDF Relevance: The three given activities, Thermometer Demonstration, Video Animation, and Case Study Presentation, are aligned with the Instructional Design Framework by integrating the four key aspects: context, connection, collaboration, and creativity. The Thermometer Demonstration provides contextual grounding through hands-on observation, helping students connect external temperature changes to internal body responses. The Video Animation enhances conceptual connection by visually illustrating the hypothalamus’s role in thermoregulation, while promoting collaborative discussion and inquiry. The Case Study Presentation fosters creativity and deeper analysis by linking clinical symptoms to endocrine function, encouraging students to interpret real-world scenarios through scientific reasoning. Pedagogically, these activities support experiential, visual, and inquiry-based learning, making abstract concepts more accessible and engaging for diverse learners.
--	---	--

- Describe how it receives signals from thermoreceptors and sends instructions to effectors (sweat glands, muscles, blood vessels)

- Reinforce the analogy:

Thermometers = sensory receptors

Hypothalamus = control center

Body = effectors

- Class discussion and diagram

- Draw a simple flowchart on the board or show it through a slide:

Code: Temperature Change → Thermoreceptors → Hypothalamus → Nervous/Endocrine Response → Effectors

- Use student input to label each part and describe the response.

Guiding Questions:

- What did the thermometer readings tell us about the environment?
- How does your skin detect similar changes in temperature?
- What role does the hypothalamus play in responding to these changes?
- Why do we sweat when it's hot and shiver when it's cold?
- How do blood vessels help regulate body temperature?
- Can you think of a situation where both the nervous and endocrine systems work together to maintain balance?

Option 2: Video Animation

(https://www.youtube.com/watch?v=ac_8bjEZaVE)

Procedure:

- The teacher plays the video showing the segment where the hypothalamus is introduced as the body's thermostat. The video explains how it compares body temperature to a set point and initiates cooling or warming responses through blood vessels, shivering, and hormonal release.

- The teacher might strategically pause the video for inquiry when the hypothalamus is shown receiving temperature signals and issuing commands. This is a natural moment to prompt reflection and discussion.
- Ask the students about the post-guiding questions and encourage students to refer to what they saw (e.g., blood vessel constriction, shivering, hormone release):
 - o What signals or responses are initiated by the body to regulate temperature and maintain homeostasis?
 - o How do the different signals and responses work together to regulate body temperature, and what might happen if one of these processes fails?
- The teacher must facilitate a brief discussion by letting the students share responses. Reinforce the role of the hypothalamus as both a sensor and regulator, integrating signals and triggering nervous and endocrine responses.
- To end, the teacher must bridge to the lesson objective with the prompt: *“Now that we’ve seen how the hypothalamus works, let’s explore how it coordinates with the nervous and endocrine systems to keep your body in balance.”*

Option 3: Case Study Presentation (*Hyperthyroidism and Heat Sensitivity*)

Procedure:

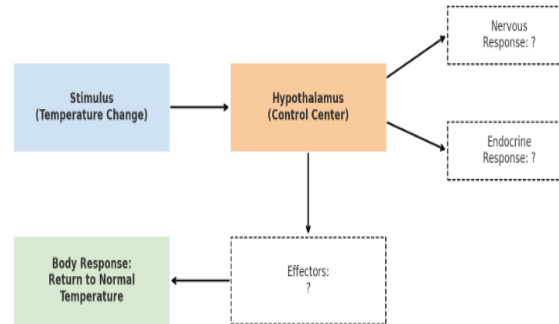
- The teacher may introduce a brief clinical scenario:
“Meet Daniella, a 16-year-old student who often feels overheated, even in cool environments. She sweats excessively, feels anxious, and has a rapid heartbeat. Her doctor diagnosed her with hyperthyroidism.”
 Optionally, the teacher may display a simple infographic or short video clip showing symptoms and effects of thyroid imbalance.

	- Ask the students the following questions to facilitate guided analysis: • <i>Why do you think Daniella feels unusually hot despite being in a cool environment?</i> • <i>What role do thyroid hormones play in regulating body temperature and metabolism?</i> • <i>How might excessive thyroid hormone production lead to the symptoms Daniella is experiencing, such as sweating, anxiety, and rapid heartbeat?</i> • <i>What could happen to the balance of her body systems if hyperthyroidism is left untreated?</i> - Encourage students to connect symptoms to internal processes, such as increased metabolism and heat production. - The teacher introduces a scientific explanation by clarifying that the thyroid hormones (T3 and T4) regulate metabolism and can accelerate cellular activity if in excess, which can lead to an increase in heat generation. <i>(Note: Link this to the concept of thermoregulation and how endocrine disorders disrupt it)</i> - Conclude the activity by reinforcing that the endocrine system, particularly the thyroid gland, plays a critical role in maintaining body temperature. Also, the teacher may highlight how disorders like hyperthyroidism or hypothyroidism affect homeostasis and overall health. 2. Discussing New Concept Option 1. Diagram Labelling Procedure: - The teacher distributes to the students the partially completed thermoregulation diagram illustrating the thermoregulation	IDF Relevance: The three given activities, Diagram Labelling, PHET Simulation, and Storytelling, strongly reflect the Instructional Design Framework by integrating the four key aspects: context, connection, collaboration, and creativity. Diagram Labelling provides contextual grounding by visually mapping the
--	---	--

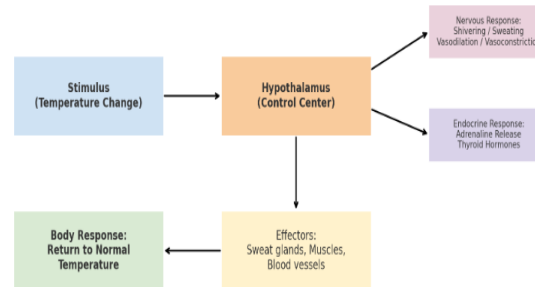
process (e.g., stimulus → receptor → control center → effector → response).

- The teacher guides the students in labeling key components of the diagram.
- The teacher prompts the questions:
 - “Where does the signal go next?” or
 - “What part of the body responds to the signal?”
- To end, reinforce the flow of information and feedback loop between the nervous and endocrine systems.

Partially completed diagram



Completed Diagram



Expected Output: Accurate labeled diagram.

thermoregulation process, helping students connect abstract concepts to concrete biological functions through guided questioning. The PHET Simulation enhances connection and collaboration as learners actively explore how environmental and behavioral changes affect internal body responses, encouraging inquiry, peer dialogue, and pattern recognition. Storytelling fosters creativity and embodied learning by transforming physiological roles into relatable characters, allowing students to internalize system coordination through roleplay and narrative. Pedagogically, these activities support multimodal, constructivist, and inclusive approaches that promote deeper understanding, active engagement, and meaningful integration of science concepts across diverse learner profiles.

	Option 2: Simulation Activity: Exploring Thermoregulation through PHET PhET Interactive Simulation – Eating and Exercise (https://phet.colorado.edu/en/simulations/eating-and-exercise) Procedure: - The teacher projects the simulation on the screen. Introduce the activity by explaining that students will explore how the body responds to changes in internal and external conditions. - Let the students manipulate the PhET by adjusting the environmental temperature or the activity levels (e.g., exercise intensity) to simulate heat production. Observe how the virtual body responds (e.g., sweating, increased heart rate, or changes in energy use). - Allow the students to record their observations of the students in both heat and cold conditions in a two-column table. Prompts may include: ○ <i>“What changes occur when the body is exposed to heat?”</i> ○ <i>“How does the body respond to cold?”</i> ○ <i>“Which systems seem to be involved?”</i> - The teacher facilitates discussion and debriefing. Guide the students in comparing their notes and identifying patterns. Highlight the roles of the nervous system (e.g., shivering, vasoconstriction) and endocrine system (e.g., hormone release) Expected Output: The students must produce a structured observation note comparing body responses to heat and cold, and begin to understand the coordinated role of internal systems in thermoregulation.	Note to Teacher: While PhET doesn’t have a direct “thermoregulation” simulation, the Eating & Exercise module allows students to explore metabolic responses, which can be adapted to discuss heat generation and regulation. In this activity, the teacher will act as facilitator in manipulating the PhET simulation.
--	--	---

	Option 3: Storytelling – “A Day in the Life of a Hypothalamus” Procedure: - The teacher narrates a lively story where the hypothalamus is portrayed as “the king” of body temperature control, receiving signals from receptors and issuing commands to glands and muscles. - The students take on roles such as temperature receptors, hypothalamus, sweat glands, and muscles, acting out the sequence of events when the body gets too hot or too cold. - Include simple props or gestures (e.g., fans for sweating, shivering motions) to reinforce the responses. Expected Output: The students demonstrate the thermoregulation process through roleplay, showing how the nervous and endocrine systems coordinate to maintain balance. Script: English Version Title: <i>A Day in the Life of a Hypothalamus</i> Narrator Introduction Welcome to the control room of the human body! Meet the Hypothalamus, the king of body temperature. <u>Scene 1: The Heat Alert</u> Narrator: It’s a hot afternoon. The skin’s temperature receptors send a message to the hypothalamus. Receptor (student): King, it’s getting hot out here! Messenger Neuron (student): Message delivered!	Note to Teacher: This activity uses narrative and roleplay to make the abstract concept of thermoregulation more concrete and engaging. By personifying the hypothalamus as “the king” and assigning students roles such as receptors, messenger neurons, sweat glands, and muscles, learners actively explore how the nervous and endocrine systems coordinate to maintain body temperature. As the facilitator of learning, encourage expressive participation and use simple props or gestures to reinforce key actions (e.g., shivering, sweating). The bilingual script provided supports inclusivity and accessibility, especially in multilingual classrooms. Also, the script can be elaborated further suitable to the topic being learned.
--	--	--

Hypothalamus (student): Time to cool down! Sweat glands, activate!
Blood vessels, widen!

Sweat Glands (student): Releasing sweat now!

Blood Vessels (student): Expanding to release heat!

Scene 2: The Cold Crisis

Narrator: Later that night, the temperature drops. Receptors send another alert.

Receptor (student): King, it's freezing out here!

Messenger Neuron (student): Message delivered again!

Hypothalamus (student): Time to warm up! Muscles, start shivering!
Blood vessels, tighten up!

Muscles (student): Shivering in progress!

Blood Vessels (student): Narrowing to keep heat in!

Closing Line

Narrator: And that's how the hypothalamus keeps your body just right, not too hot, not too cold.

Script: Tagalog Version

Pamagat: ***Isang Araw sa Buhay ng Hypothalamus***

Panimula ng Tagapagsalaysay

Maligayang pagdating sa control room ng katawan ng tao! Kilalanin natin ang Hypothalamus, ang hari ng temperatura ng katawan.

Eksena 1: Babala ng Init

Tagapagsalaysay: Mainit ang hapon. Nagpadala ng mensahe ang mga temperature receptor sa balat papunta sa hypothalamus.

Receptor (mag-aaral): Hari, umiinit na dito!

Neuron na Mensahero (mag-aaral): Mensahe naihatid!

	Hypothalamus (mag-aaral): Panahon nang magpalamig! Sweat glands, kumilos! Blood vessels, lumuwag! Sweat Glands (mag-aaral): Naglalabas na kami ng pawis! Blood Vessels (mag-aaral): Lumalawak para maglabas ng init! <u>Eksena 2: Krisis ng Lamig</u> Tagapagsalaysay: Kinagabihan, bumaba ang temperatura. Nagpadala ulit ng alerto ang mga receptor. Receptor (mag-aaral): Hari, sobrang lamig dito! Neuron na Mensahero (mag-aaral): Mensahe naihatid ulit! Hypothalamus (mag-aaral): Panahon nang magpainit! Muscles, magsimulang manginig! Blood vessels, maghigpit! Muscles (mag-aaral): Nanginig na kami! Blood Vessels (mag-aaral): Naghihigpit para mapanatili ang init! <u>Pangwakas na Linya</u> Tagapagsalaysay: At ganyan kung paano pinapanatili ng hypothalamus ang tamang temperatura ng katawan, hindi masyadong mainit, hindi masyadong malamig. 3. Developing Mastery Option 1: Thermoregulation in Action: A Roleplay Procedure: - The teacher divides the class into smaller groups. Each group selects or is assigned a particular scenario (e.g., running under the sun, jumping into cold water, and walking in smoggy heat) - The students take on the following roles: • Sensory Receptors (e.g., skin detecting temperature) • Messenger Neurons (delivering signals) • Hypothalamus (the control center)	IDF Relevance: The Thermoregulation Roleplay, Concept Mastery Game via Quizizz/Kahoot, and the Pathway Worksheet are thoughtfully designed learning experiences that exemplify the DepEd Instructional Design Framework by embedding the principles of context, connection, collaboration, and creativity. Through roleplay, students engage in experiential learning by dramatizing real-life temperature scenarios, which strengthens their understanding of system coordination using movement and imagination. The interactive quiz game fosters active recall
--	--	---

- Effectors (e.g., sweat glands, muscles, blood vessels)
- The groups perform a short skit showing the correct thermoregulation sequence: stimulus → receptor → control center → effector → response.
- The teacher observes and provides feedback on the accuracy, clarity, and creativity of each group in performing each role.

Expected Output: The student-led skits that correctly illustrate the thermoregulation pathway, showing coordination between body systems.

Option 2: Concept Mastery Game Through Quizizz/Kahoot

Procedure:

- The teacher launches a 10-item interactive quiz using Quizizz or Kahoot, covering concepts such as:
 - Roles of the hypothalamus
 - Differences between nervous and endocrine responses
 - Examples of effectors in heat/cold scenarios
 - Hormonal vs. electrical signaling
- The students respond in real time using their devices.
- The teacher should provide immediate feedback after the activity, reinforcing correct answers and addressing misconceptions.

❖ *Note: The teacher may use the following questions for Quizizz/Kahoot*

1. What is the primary role of the hypothalamus in thermoregulation?

- A. To produce hormones C. To regulate blood pressure
B. To act as the body's thermostat D. To control digestion

Correct answer: B

2. Which system uses electrical signals for rapid communication?

- A. Digestive system C. Nervous system
B. Circulatory system D. Endocrine system

Correct answer: C

and peer interaction through ICT integration, while immediate feedback reinforces key concepts. Meanwhile, the pathway worksheet promotes visual mapping and systems thinking, guiding students to collaboratively trace physiological responses to cold. These activities employ constructivist, gamified, and differentiated pedagogical approaches that support inclusive, engaging, and mastery-focused science instruction.

Option 1 Note to Teacher:

The teacher may use the provided Rubric to evaluate the performance of the students in their Roleplay.

Criteria	4 – Exemplary	3 – Proficient	2 – Developing	1 – Beginning
Scientific Accuracy	All components of the thermoregulation pathway are correctly represented and clearly explained.	Most components are accurate; minor errors do not hinder understanding.	Some components are correct; key concepts may be missing or unclear.	Major misconceptions or missing steps in the pathway.
Role Clarity & Coordination	Each group member clearly embodies their assigned role (e.g., receptor, hypothalamus, effector) and transitions smoothly.	Roles are mostly clear; transitions are generally smooth.	Roles are somewhat unclear; coordination needs improvement.	Roles are confusing or incomplete; group lacks coordination.
Communication & Expression	Dialogue is clear, confident, and engaging; scientific terms are used appropriately.	Dialogue is mostly clear; scientific terms are used with some accuracy.	Dialogue is basic; limited use of scientific vocabulary.	Dialogue is unclear or off-topic; minimal scientific language is used.

Which system uses hormones to regulate body functions over time?

- A. Nervous system C. Respiratory system
B. Endocrine system D. Immune system

Correct answer: B

3. **What is an example of an effector in response to heat?**

- A. Hair follicles C. Adrenal glands
B. Sweat glands D. Liver

Correct answer: B

4. **Which of the following is an effector response to cold exposure?**

- A. Sweating C. Shivering
B. Vasodilation D. Increased heart rate

Correct answer: C

5. **How does the nervous system differ from the endocrine system in terms of response speed?**

- A. It is faster and short-lived C. It is slower and short-lived
B. It is slower and long-lived D. It is faster and long-lived

Correct answer: A

6. **Which hormone is involved in increasing metabolism and heat production?**

- A. Insulin C. Melatonin
B. Thyroxine D. Oxytocin

Correct answer: B

7. **What type of signal does the endocrine system use?**

- A. Electrical impulses C. Neural signals
B. Chemical hormones D. Mechanical signals

Correct answer: B

8. **Which of the following best describes the interaction between the nervous and endocrine systems?**

- A. They work independently
B. They are antagonistic
C. They work together to maintain homeostasis
D. They regulate only voluntary actions

Correct answer: C

9. **What happens to blood vessels during heat exposure?**

- A. They contract to conserve heat C. They dilate to release heat
B. They constrict to slow blood flow D. They break down

Correct answer: C

Creativity & Engagement	Skit is imaginative, well-paced, and captures audience's attention.	Skit is organized and moderately engaging.	Skit shows effort but lacks energy or creativity.	Skit is disorganized or lacks engagement.
Understanding of System Coordination	Demonstrates strong grasp of how nervous and endocrine systems interact to regulate temperature.	Shows general understanding of system interaction.	Partial understanding; systems may be treated separately.	Limited or no understanding of system coordination.
Total score:				

Option 3: Pathway Worksheet: Tracing the Cold Response

Procedure:

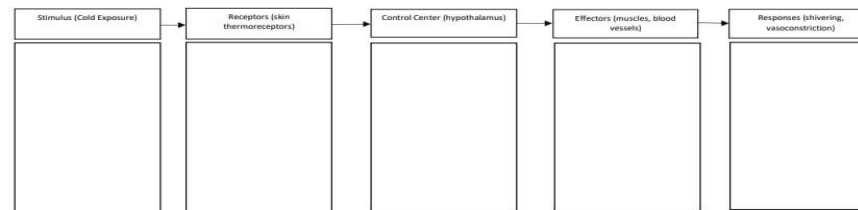
- The teacher divides the class into groups. Each group receives a worksheet with a scenario: *"You jump into cold water, what happens next?"*
- The students trace the stimulus-response pathway by identifying:
 - Stimulus (cold exposure)
 - Receptors (skin thermoreceptors)
 - Control center (hypothalamus)
 - Effectors (muscles, blood vessels)
 - Responses (shivering, vasoconstriction)
- Allow the students to work on the worksheet to encourage visual organization and systems thinking.
- The teacher then checks for accuracy and provides feedback or peer review.

Worksheet for the activity

Tracing the Cold Response: Thermoregulation Pathway

You jump into cold water—what happens next?

Trace the pathway your body follows to respond to cold. Fill in each box with explanations or illustrations.



Teachers Feedback/Peer Review Notes:

--

	Expected Output: Completed flowcharts of the students showing correct sequence and system coordination in response to cold exposure.	
C. Demonstrating Knowledge and Skills	1. Finding Practical Application Option 1: Scenario Cards – System Response Mapping Procedure: - The teacher prepares the scenario cards featuring real-life situations such as, but not limited to, an athlete finishing a marathon, a child with a fever, an elderly person in cold weather, and a student giving a speech. - Divide the students into small groups and distribute one card per group. - The teacher instructs the groups to: • Identify the stimulus in the scenario. • Trace the nervous system response (example: hormone release, effect on the target organ) • Trace the endocrine system response (example: hormone release, effect on the target organ) - Guide the students to organize their findings into a flowchart that shows how both systems work together to maintain homeostasis. Expected Output: A flowchart explaining the sequence of responses of the systems.	IDF Relevance: The activities Scenario Cards, Thermoregulation Flowchart Challenge, and Interview Assignment serve as dynamic platforms for applying scientific understanding in practical, real-world contexts, aligned with the Instructional Design Framework. The context is integrated by anchoring learning in everyday situations, fostering connection by linking physiological processes to observable responses, encouraging collaboration through group tasks and peer interaction, and promoting creativity via visual models and personalized interviews. Pedagogically, these activities utilize inquiry-based, constructivist, and differentiated approaches that support relevance, engagement, and deeper mastery of system coordination in thermoregulation.

Option 2. Thermoregulation Model: A Group Challenge Flowchart

Procedure:

- The teacher introduces the concept of Thermoregulation and how both systems respond to temperature changes.
- Assign groups to create a flowchart for either HOT or COLD conditions.
- Provide guiding questions:
 - What triggers the response (example: the skin receptors)
 - What role does the hypothalamus play?
 - Which hormones are released?
 - What are the physiological effects?
- To conclude, encourage the use of color coding or icons to distinguish nervous vs. endocrine actions

Expected Output: A visual model with annotations showing system coordination.

Option 3: Interview Assignment – Community Coping Strategies

Procedure:

- The teacher briefs the students on the activity goal: connecting science to lived experiences.
- Assign students to interview a family member or neighbor about how they cope with **extreme cold or heat**.
- The teacher provides a simple interview guide about the scenario:
 - What strategies do they use (e.g., hydration, clothing, rest)?
 - How do these help the body respond?
 - Are there cultural or traditional practices involved?

	- The teacher asks the students to link these strategies to the nervous and endocrine system functions. Expected Output: A written transcript of the interview with student annotations explaining the science behind each response. 2. Making Generalization Option 1: System Integration through Group Discussion Procedure: - The teacher writes or projects the guiding question on the board: <i>“Why is integration of both the nervous and endocrine systems essential for survival?”</i> - Briefly review how each system functions individually and how they complement each other (example: speed vs. duration of response) - Divide students into small groups (3-5 members, depending on the class size) - Ask each group to discuss the question for 5-7 minutes, using examples from previous activities (example: thermoregulation, stress response) - Encourage the students to consider the following: • What would happen if only one system responded? • How do they work together to maintain homeostasis? - Request each group to share a summary of their discussion with the class. Expected Output: Collective answers or a summary statement of the discussion conducted. Option 2: Concept Mapping – Comparing System Roles	IDF Relevance: These three activities, System Integration through Group Discussion, Concept Mapping, and Hypothalamus Reflection Exit Ticket, embody the DepEd Instructional Design Framework by integrating the four key aspects: context, connection, collaboration, and creativity. The group discussion anchors learning in real-life physiological scenarios, fostering contextual understanding and collaborative exchange of ideas. Concept mapping encourages students to creatively organize and connect concepts across body systems, deepening their grasp of regulation mechanisms. The exit ticket promotes reflective thinking by linking classroom content to personal insights, reinforcing meaningful connections. Together, these tasks support inclusive, learner-centered instruction that builds scientific understanding through active engagement and imaginative expression. Note to Teacher: For option 1. The teacher may record key ideas on the board to synthesize class understanding.
--	--	---

	Procedure: - The teacher instructs the students to create their own concept map. - Let the students create a visual comparison of the Nervous system and Endocrine system contributions to body regulation. - Guide the students to include: • Key components (e.g., brain, spinal cord, glands, hormones) • Types of stimuli and responses • Speed and duration of action • Example of coordinated functions (e.g., fight-or-flight response, temperature control) - Allow the students to work individually or in pairs, depending on time and class size. Expected Output: A completed map showing distinctions and overlaps between the two systems. Option 3: Hypothalamus Reflection: An Exit Ticket Procedure: - The teacher distributes a small slip of paper or uses the lecture notebooks of the students for a quick written reflection. - The teacher prompts: “The hypothalamus is important because...?” - Ask the students to write 1-2 sentences explaining the role of the hypothalamus in coordinating the nervous and endocrine responses. - Encourage the students to use specific examples like temperature regulation, hormone release, among other relevant responses. Expected Output: Submitted written reflections.	For Option 2. The teacher may intend to have a gallery walk or peer review feedback on the outputs. IDF Relevance: These three activities, Survival Kit Designing, Case Analysis of Hypothalamic Dysfunction, and Body Systems Emergency Briefing, align with the
--	--	--

	3. Evaluating Learning Option 1: Survival Kit Designing for Extreme Temperatures <i>The goal of this activity is to apply the students' understanding of nervous and endocrine system responses by designing a survival kit tailored for either extreme heat or extreme cold, explaining the science behind each item.</i> Materials: - Manila paper or cartolina (for posters) - Markers, crayons, pens, sticky notes - Printed pictures/magazines (optional for visuals) - Scissors, tape, glue - Optional: digital tools (PowerPoint, Canva, or Google Slides for group presentations) Procedure: - Present this scenario: <i>“Imagine you are preparing a survival kit for someone living in a desert or a snowy mountain. What items would you include to help the body maintain homeostasis?”</i> - Divide the class into small groups (4–6 members). ▪ Assign or let each group choose one environment: Scorching HOT (desert) or Extremely COLD (snowy mountain). - Each group discusses and lists: ▪ Physiological challenges in their chosen environment (e.g., dehydration, heatstroke, frostbite, hypothermia). ▪ Body responses through the nervous and endocrine systems (e.g., sweating, vasodilation, shivering, hormone release). ▪ Survival items that can support the body (e.g., water bottles, layered clothing, high-energy food). - For each item in the survival kit, groups must explain: ▪ What body system does it support? ▪ What hormone or physiological response does it relate to (e.g., adrenaline for heat stress, thyroid hormone for metabolism, shivering for warmth)?	Instructional Design Framework. Each task is rooted in real-life scenarios that provide meaningful context for understanding thermoregulation. Students build connections between scientific concepts and practical applications, such as survival strategies and medical symptoms. Collaboration is fostered through group discussions, case analysis, and peer presentations, encouraging shared inquiry and teamwork. Creativity is expressed through designing kits, crafting visual outputs, and simulating emergency briefings, allowing learners to demonstrate understanding in diverse and imaginative ways. These activities promote deeper learning by engaging students in purposeful, connected, and collaborative exploration of body system integration. Note to teacher: For option 1. The teacher may opt to have a peer review or a gallery walk to compare the models. Also, encourage creativity (illustrations, models, or culturally relevant survival items). For option 2. The teacher must review the responses of the students to assess their understanding and to identify any misconceptions of the lesson.
--	---	---

- How does it help maintain homeostasis?
- Groups prepare a poster, slide deck, or physical mock-up of their survival kit.
- Facilitate a discussion comparing kits from hot vs. cold environments.
- Highlight similarities and differences in body responses and survival strategies.

Guiding Questions:

- Why is water more important in hot environments compared to cold ones?
- How does clothing choice (light vs. layered) affect thermoregulation?
- Which hormones are most active in helping the body adapt to extreme temperatures?
- What could happen if the survival kit lacked one of these key items?
- How do the nervous and endocrine systems work together to keep the body balanced in these conditions?

Expected Output: Survival kit list with scientific justification

Option 2: Case Analysis: The Malfunctioning Thermostat”

The goal of this activity is to apply the understanding of nervous and endocrine system integration in thermoregulation by analyzing a case of hypothalamic dysfunction.

Procedure:

- Present the sample case to the students:

Case Title: Miguel’s Mystery Symptoms

Miguel, a 16-year-old student, has recently been experiencing strange symptoms. Even on hot days, he doesn’t sweat and often feels cold. His body shivers uncontrollably, even when others are comfortable.

Sample Rubric for the Survival Kit Designing

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)
Scientific Accuracy				
Creativity and Presentation				
Team Collaboration				
Explanation of Body Responses				
Application of Concepts				

Sample Rubric for the Case Analysis

Criteria	Excellent (3)	Good (2)	Needs Improvement (1)
Understanding of Systems	Clearly explains how the nervous and endocrine systems work together	Explains basic roles of both systems	Shows confusion or misses key ideas
Use of Case Details	Uses specific symptoms and links them to system functions	Mentions some symptoms with partial explanation	Few or no connections to the case
Explanation of Hypothalamus Role	Accurately describes its function in thermoregulation	Mentions the hypothalamus with some detail	Limited or unclear explanation
Clarity of Response	Well-organized and easy to follow	Mostly clear with minor issues	Hard to follow or incomplete

The doctors found that his thyroid hormone levels are usually low, and his adrenal glands aren't releasing adrenaline properly. They suspect a problem with Miguel's hypothalamus.

Miguel also reports feeling tired, dizzy, and unable to concentrate. His skin sometimes appears pale, and his heart remains low even during physical activity.

- In pairs or small groups, analyze the case by answering the following prompts:
 - What symptoms suggest a problem with thermoregulation?
 - Which body systems are involved?
 - What is the role of the hypothalamus in this case?
 - How might the nervous and endocrine systems be failing to coordinate?
 - What are the possible consequences for Miguel's homeostasis?
 - Suggest basic interventions or coping strategies Miguel could use.
- The students should create a response sheet that includes a short explanation (in paragraph or in bullet form) and a labeled diagram or flowchart showing pathways.

Expected Output: A response sheet and a labeled diagram or flowchart.

Option 3: Make a Scene: Body Systems Emergency Briefing

The goal of the activity is for the students to explain how the body responds to extreme heat or cold using scientific terms.

Procedure:

- The teacher sets up a scene and tells the students:

Total Score:	
--------------	--

***Sample Rubric for the Body Systems
Emergency Briefing***

Criteria	Excellent (3)	Good (2)	Needs Improvement (1)
Scientific Content Accuracy	All concepts are clearly explained using correct scientific terms and real-life examples.	Most concepts are accurate with minor errors or missing details.	Some concepts are present but lack clarity or contain noticeable inaccuracies.
Explanation of Body Responses	Thorough explanation of nervous and endocrine roles, the hypothalamus function, and organ responses.	Adequate explanation with some connections between systems and body responses.	Basic explanation with limited integration of systems and responses.
Problem Analysis & Safety Tips	Insightful discussion of potential issues and well-justified safety recommendations.	Good discussion with relevant safety tips and basic reasoning.	Limited discussion with general or vague safety suggestions.
Communication & Presentation	Clear, confident delivery with organized visuals or handouts; effectively engages the audience.	The presentation is organized and understandable, with some visual support.	Presentation is somewhat disorganized or lacks clarity; minimal visual support.
Team Collaboration	All members contribute	Most members participate; the	Uneven participation;

	meaningfully; strong evidence of teamwork and shared responsibility.	group works well together.	some collaboration issues.
Total Score:			

“Imagine you are part of a team advising people who will travel to extremely hot or cold environments. You need to explain how the body maintains balance, or homeostasis, and what processes help it adjust to these conditions.”

Divide the class into small groups and let the students prepare a short presentation or poster that includes:

- How do the nervous and endocrine systems help with temperature control?
- What does the hypothalamus do?
- What happens to sweat glands, muscles, and blood vessels
- What can go wrong (e.g., no sweating, too much shivering)
- Tips to stay safe (e.g., drinking water, wearing layers of clothing)

- Let each group present its work to the class.

Expected Output: A poster, short talk, or handout using simple science terms and real-life examples.

4. Additional Activities

Option 1. Mini-Research – Climate Change and Thermoregulation

Procedure:

- The teacher introduces how climate change affects body temperature regulation.
Note: This article (*J Clin Invest. 2020;130(2):559–561. <https://doi.org/10.1172/JCI135006>*) can be used as a reference.
- Guide the students to research using the following prompt questions:

IDF Relevance:

These enrichment activities, Mini-Research on Climate Change and Thermoregulation, and Science Journal Personal Reflection, are grounded in the 5E Instructional Design Framework and promote inquiry-based, learner-centered pedagogy. They begin by engaging students through personal or global contexts that make thermoregulation relevant and relatable. Students explore scientific challenges posed by extreme temperatures and climate change, then explain the physiological roles of the nervous and endocrine systems using real-world examples. Through elaboration, they connect abstract concepts to lived experiences or societal issues, fostering deeper understanding. The evaluation phase encourages metacognition and synthesis through written reflections or research summaries. Pedagogically, these tasks support differentiated instruction, critical thinking, and constructivist learning by allowing students to build

- In what ways does climate change make it harder for the human body to maintain a stable internal temperature?
 - How do the nervous and endocrine systems respond to these challenges?
- Provide sources or let the students search with support.
 - Students write a summary with one real-world example and a suggested solution.

Expected Output: A short, written explanation of how climate change leads to more extreme temperatures and how these conditions challenge the body's ability to regulate temperature.

Option 2. Science Journal – Personal Reflection

Procedure:

- The teacher asks the students to recall a time they felt too hot or too cold.
- Guide the students to describe what their body did in those conditions (e.g., sweating, shivering)
- The students write a short journal entry (5-7 sentences) linking their experiences to body systems.

Expected Output: A journal entry that includes a personal experience of the students in too hot or too cold conditions, describing their physical sensations and explaining how their body responds to it.

These are enrichment, reinforcement, or remediation activities designed to support diverse learners. These can be extension work, research tasks, or differentiated exercises.

knowledge through meaningful exploration and personal expression.

Sample Rubric for the Mini Research

Criteria	Excellent (3)	Good (2)	Needs Improvement (1)
Scientific Understanding	Clearly explains how climate change affects thermoregulation using accurate scientific terms and concepts.	Explains the topic with mostly accurate science; minor errors or gaps.	Basic explanation with limited scientific depth or clarity.
Systems Integration	Effectively describes how both nervous and endocrine systems respond to temperature stress.	Describes both systems with some integration; may lack depth or clarity.	Mentions systems but with minimal connection or detail.
Real-World Example	Provides a relevant, well-explained example that illustrates the impact of climate change on human thermoregulation.	The example is relevant but lacks a full explanation or connection to the topic.	The example is vague or loosely connected to thermoregulation.
Suggested Solution	Offers a thoughtful, science-based solution to mitigate thermoregulatory challenges.	Suggests a reasonable solution with basic scientific support.	The suggestion is general or lacks scientific justification.

			<table><tr><td>Clarity and Organization</td><td>Writing is clear, well-organized, and free of major errors; ideas flow logically.</td><td>Mostly clear and organized; minor issues with flow or grammar.</td><td>Writing is somewhat disorganized or unclear; multiple errors are present.</td></tr><tr><td colspan="4">Total Score:</td></tr></table>	Clarity and Organization	Writing is clear, well-organized, and free of major errors; ideas flow logically.	Mostly clear and organized; minor issues with flow or grammar.	Writing is somewhat disorganized or unclear; multiple errors are present.	Total Score:																			
			Clarity and Organization	Writing is clear, well-organized, and free of major errors; ideas flow logically.	Mostly clear and organized; minor issues with flow or grammar.	Writing is somewhat disorganized or unclear; multiple errors are present.																					
			Total Score:																								
			Sample Rubric for the Science Journal																								
			<table><tr><th>Criteria</th><th>Excellent (3)</th><th>Good (2)</th><th>Needs Improvement (1)</th></tr><tr><td>Personal Experience</td><td>Clearly describes a specific moment of feeling too hot or cold with vivid detail.</td><td>Describes a relevant experience with some detail.</td><td>Mentions an experience but lacks clarity or depth.</td></tr><tr><td>Physical Sensations</td><td>Accurately describes body responses (e.g., sweating, shivering) with a clear connection to the experience.</td><td>Describes body responses with basic accuracy.</td><td>Mentions body responses but lacks explanation.</td></tr><tr><td>Scientific Explanation</td><td>Effectively links body responses to nervous and endocrine systems using correct terms.</td><td>Makes basic connections to body systems with minor errors.</td><td>Attempts to link to body systems but lacks clarity or accuracy.</td></tr><tr><td>Clarity and Organization</td><td>Writing is well-structured, easy to follow, and free of major errors.</td><td>Mostly clear and organized; minor grammar or flow issues.</td><td>There are some organizational issues, multiple grammar or clarity problems.</td></tr><tr><td colspan="4">Total Score:</td></tr></table>	Criteria	Excellent (3)	Good (2)	Needs Improvement (1)	Personal Experience	Clearly describes a specific moment of feeling too hot or cold with vivid detail.	Describes a relevant experience with some detail.	Mentions an experience but lacks clarity or depth.	Physical Sensations	Accurately describes body responses (e.g., sweating, shivering) with a clear connection to the experience.	Describes body responses with basic accuracy.	Mentions body responses but lacks explanation.	Scientific Explanation	Effectively links body responses to nervous and endocrine systems using correct terms.	Makes basic connections to body systems with minor errors.	Attempts to link to body systems but lacks clarity or accuracy.	Clarity and Organization	Writing is well-structured, easy to follow, and free of major errors.	Mostly clear and organized; minor grammar or flow issues.	There are some organizational issues, multiple grammar or clarity problems.	Total Score:			
			Criteria	Excellent (3)	Good (2)	Needs Improvement (1)																					
			Personal Experience	Clearly describes a specific moment of feeling too hot or cold with vivid detail.	Describes a relevant experience with some detail.	Mentions an experience but lacks clarity or depth.																					
			Physical Sensations	Accurately describes body responses (e.g., sweating, shivering) with a clear connection to the experience.	Describes body responses with basic accuracy.	Mentions body responses but lacks explanation.																					
			Scientific Explanation	Effectively links body responses to nervous and endocrine systems using correct terms.	Makes basic connections to body systems with minor errors.	Attempts to link to body systems but lacks clarity or accuracy.																					
			Clarity and Organization	Writing is well-structured, easy to follow, and free of major errors.	Mostly clear and organized; minor grammar or flow issues.	There are some organizational issues, multiple grammar or clarity problems.																					
Total Score:																											

V. ASSESSMENT

(Assessing Learnings)

Summative test:

Administer a 10-question test (the type of test varies depending on the teacher who will administer the test) covering all the topics discussed, presented in a traditional paper-and-pencil format, a Google Forms digital quiz, or an interactive online game such as Kahoot or Wayground, formerly Quizizz.

Here's the sample test:

Topic: Thermoregulation and Neuroendocrine Integration

Type: Multiple Choice

Directions: Choose the letter of the correct answer.

- Which part of the brain integrates sensory input and coordinates body responses to maintain temperature balance?
A. Cerebellum B. Cerebrum **C. Hypothalamus** D. Medulla oblongata
- How does adrenaline contribute to survival in cold environments beyond its immediate effects?
A. Slows heart rate B. Causes vasodilation C. Stimulates sweating **D. Increases metabolic activity**
- Which hormone plays a key role in increasing metabolic rate to generate heat, and how does this support homeostasis?
A. Insulin B. Thyroid hormone **C. Estrogen** D. Cortisol
- Which of the following is NOT directly involved in executing thermoregulatory responses?
A. Sweat glands B. Muscles C. Blood vessels **D. Pancreas**
- What physiological change occurs during vasodilation, and how does it help regulate body temperature?
A. Blood vessels narrow to conserve heat C. Muscles contract to generate heat
B. Blood vessels widen to release heat D. Hormones are released to cool the body
- Which response is triggered by the nervous system to increase heat production in cold conditions?
A. Sweating B. Vasodilation **C. Shivering** D. Increased hunger
- How do sweat glands function as effectors in thermoregulation, and what would happen if they failed?
A. By releasing sweat to cool the body C. By releasing hormones
B. By constricting blood vessels D. By producing heat
- Why is it essential for the nervous and endocrine systems to work together during temperature stress?
A. It prevents sweating C. It stops hormone production
B. It increases digestion **D. It allows fast and long-term responses**
- Which system provides rapid responses to temperature changes, and how does this complement hormonal regulation?
A. Endocrine system B. Digestive system **C. Nervous system** D. Circulatory system

	10. Which response conserves heat in cold conditions, and what structural change enables this effect? <i>(Relational)</i> A. Vasodilation B. Sweating C. Vasoconstriction D. Increased urination
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	<i>The lesson reflection may include, but is not limited to:</i> Teacher Notes: ➤ Students were actively engaged in the lesson and visual activities. ➤ Some required additional scaffolding for students who cannot keep pace with the lesson (Plan for more peer-assisted strategies). ➤ Strong interest in real-life scenarios and/or experiences. Learner Feedback: ➤ Gained deeper insights into the coordination of the neuroendocrine system. ➤ Requested more real-world examples and hands-on activities to deepen the understanding of the lesson.